

Lessons Learned in Designing Racially Diverse Androgynous Avatars

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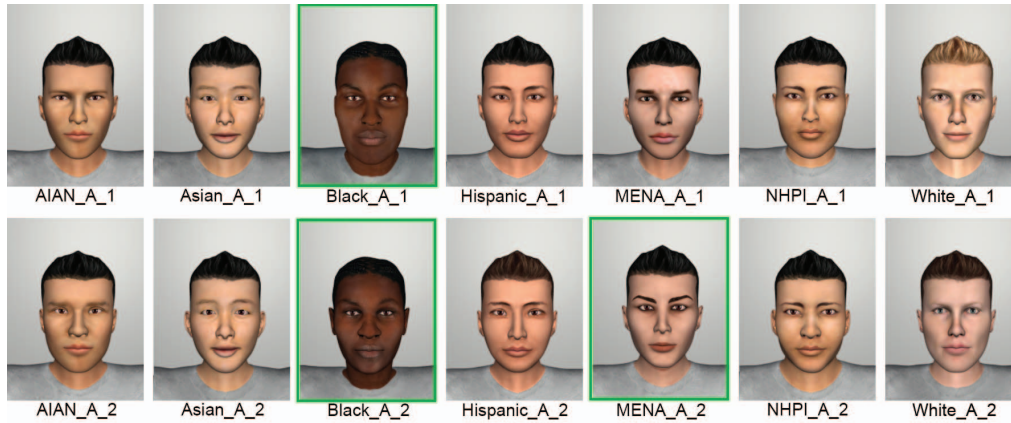


Figure 1: The 14 avatars created for our research. Avatars highlighted in green were perceived as androgynous.

ABSTRACT

As virtual reality technology evolves, avatars play a crucial role in user representation, yet options for gender-diverse individuals remain limited. The goal of this research was to develop 14 racially diverse androgynous avatars using the design guidelines recommended in prior work. However, a perceptual experiment involving 68 participants revealed unexpected results, as most avatars were perceived as predominantly masculine. Additionally, our results yielded discrepancies in perceived gender across different racial identities, as the same design process resulted in widely varying perceptions. Despite these challenges, our research highlights the importance of continued exploration to improve the creation of inclusive avatars.

Index Terms: Human-centered computing—User studies; Computing methodologies—Perception

1 INTRODUCTION

In the present day, usage and research regarding virtual reality (VR) has steadily increased. Specifically, more VR products have begun incorporating avatars, which represent the user in the virtual space. However, not all VR simulations allow for the user to select a virtual counterpart that “matches” their personal identity. As our understanding of gender identity and gender diversity continues to expand, it is necessary to create spaces, both real and virtual, that include those who identify or present as gender diverse. While someone who identifies as gender diverse may not always present as androgynous or gender-neutral, including an avatar option that appears as such allows for more options, potentially granting an underrepresented group access to a more immersive, inclusive VR experience [3]. Additionally, previous studies found that androgynous avatars can reduce gender bias and stereotypical assumptions

in virtual agents [1, 4] and improve student attitudes [1], providing further reason to provide the community with perceptually androgynous avatars. Currently, research surrounding androgynous virtual agents is still in its early stages. Through an iterative design process, Nag and Yalcin [4] presented a set of guidelines for developing androgynous human avatars by combining specific facial features from male and female avatars. Straight eyebrows (male), rounder and lower cheeks (female), a square jaw (male), fuller lips (female), absence of lip coloring (male), and swept back hair were considered androgynous features. However, these guidelines were only applied to and evaluated for White avatars.

Following these guidelines, we created 14 androgynous avatars across seven races: American Indian or Native Alaskan (AIAN), Asian, Black or African American (Black), Hispanic, Latino, or Spanish (Hispanic), Middle Eastern or North African (MENA), Native Hawaiian or Pacific Islander (NHPI), and White. Through an online user study, we tasked 68 participants with labeling each avatar’s perceived race and gender. However, our results show that with the exception of Black avatars and one MENA avatar, the remaining eleven avatars were perceived as masculine. Our results indicate that the previous design guidelines may not apply to all avatars due to differences such as participant population or modeling styles. Additionally, applying the same guidelines to avatars of different races resulted in widely varied perceptions of gender. Nevertheless, we present our findings to further the discussion and study of androgynous virtual agents. The models will be open sourced as part of the recently published Virtual Avatar Library for Inclusion and Diversity (VALID) [2] to further research on this topic.

2 ANDROGYNOUS AVATAR CREATION

We created androgynous avatars by combining certain facial features of previous male and female avatars from the VALID library [2], which includes avatars that have been perceptually validated for race and gender. We combined features from male and female faces of the same race, following the design guidelines presented by Nag and Yalcin [4]. For example, for an Asian androgynous avatar, we combined the straight eyebrows, square jaw, and absence of lip coloring from the male avatar with the rounder and lower cheeks, and fuller lips from the female avatar, and then applied a swept-back

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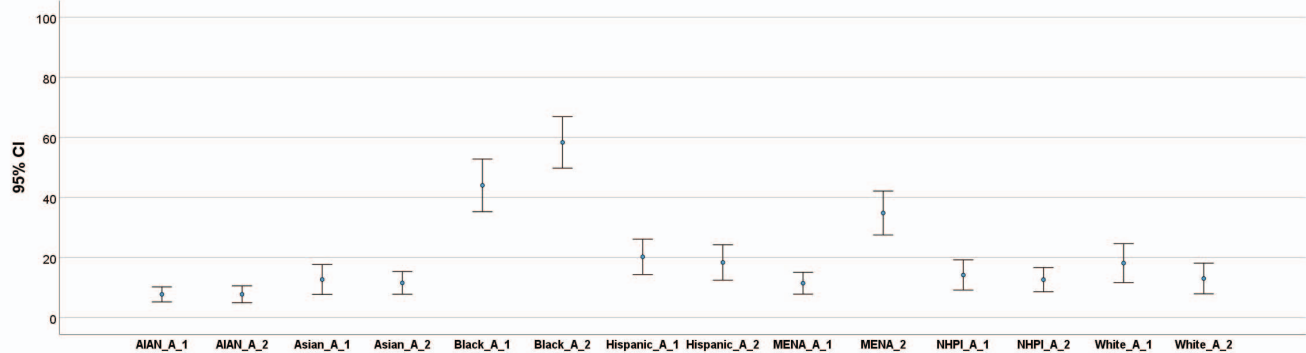


Figure 2: Plots of the 95% confidence intervals for the perceived gender of our avatars, where 0 is male and 100 is female.

hairstyle. We created a total of 14 avatars, two for each of the seven racial categories. See Figure 1 for images of the avatars created.

3 PERCEPTUAL EXPERIMENT

3.1 Procedure

We followed a procedure similar to the one used to create the original VALID avatars [2]. Participants gave their informed consent prior to taking the survey. Before taking the survey, participants were presented with definitions of the seven different racial identities (AIAN, Asian, Black, Hispanic, MENA, NHPI, and White). Next, they were asked to identify their own demographics, including age, gender, and ethnicity. Then, they viewed and classified the 14 different androgynous avatars. For each avatar, participants were shown three perspectives: a 45° left headshot, a direct or 0° headshot, and a 45° right headshot. For each avatar, participants responded to two questions: a “select all that apply” multiple-choice question asking them to identify the avatar’s race, and a slider question to identify whether the avatar was more masculine, androgynous, or feminine (similar to the procedure by Nag and Yalcin [4]). The slider’s position would start in the center between “Male” and “Female” anchors (i.e., androgynous). Participants would then move the slider to indicate their perception of the avatar’s gender. Placing the slider more toward the left identified the avatar as more masculine, while placing it toward the right identified it as more feminine.

3.2 Participants

We recruited 68 participants (33 men, 33 women, 2 non-binary) from the online Prolific marketplace to take part in the study. Participants were aged 19-40 ($M=30.26$, $SD=9.90$) with the following racial demographics breakdown: 10 AIAN, 10 Asian, 10 Black, 10 Hispanic, 10 MENA, 10 White, and 8 NHPI.

3.3 Results

To determine validated race labels for the avatars, we used the approach from the original VALID research [2]. Similar to that study, we found that participants strongly identified our Asian, Black, and White avatars as such. However, the race of our other avatars were considered ambiguous. Similar to Nag and Yalcin’s approach for androgynous avatars [4], we consider avatars to be androgynous if the 95% confidence intervals of the mean are near the mid-way point between masculine and feminine. Figure 2 shows all avatars and their gender ratings by all 68 participants.

4 DISCUSSION

Our results demonstrate that our design process was unsuccessful in universally producing avatars that were validated as androgynous, and that our avatars were mostly perceived as masculine. Unlike

Nag and Yalcin [4], who used Adobe Fuse, our 14 models were created using Autodesk Character Generator. It is possible that the facial features and other settings provided in Autodesk Character Generator may typically be perceived as more masculine. Another notable difference that could provide an explanation for our results is the difference in stimuli. Nag and Yalcin’s experiment used different lighting and did not provide side angles of the face, which also could explain our results.

Interestingly, our study had notable exceptions: both Black avatars and one MENA avatar were perceived as perceptually androgynous. These exceptions highlight the nuanced nature of perceived androgyny, suggesting potential racial influences on perceptions of facial features. Our results suggest that designers might need tailored approaches for diverse racial groups. For instance, the blend of male and female facial characteristics that creates a sense of androgyny might differ among various racial groups. This insight emphasizes the necessity for race-specific design considerations in crafting inclusive avatars, marking a crucial step towards authentic representation in virtual environments.

5 CONCLUSION

As we continue to research advancements in VR technology and broaden societal perspectives of gender identity, more research will be necessary in order to understand how to best create androgynous avatars. There are currently few investigations surrounding gender-neutral or androgynous virtual agents. While our findings revealed that the guidelines present by Nag and Yalcin were mostly not successful in producing 14 perceptually androgynous avatars, they demonstrate a need for further research in order to improve androgynous avatar creation. Nevertheless, we plan to share the 14 models for the VR community to use. Ultimately, we strive for our research to continue paving the way for future exploration and discussion regarding more inclusive virtual agents.

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